HAYOUNG KIM

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EDUCATION

Korea Cyber University — Seoul, South Korea

Bachelor of Engineering in Software Engineering — Graduated Feb 2026

Double Major in Electrical and Electronic Engineering | GPA: 4.24 / 4.50

- Areas of focus: Embedded Systems, System Software, and Computer Architecture
- Related coursework: Embedded Systems, Microprocessors, Computer Architecture, Operating Systems, and Digital Logic Design

ACTIVITIES

Fabless Jump-Up Program | Korea Fabless Industry Association — Pangyo, South Korea | Aug - Sep 2026

Telechips

- Configured a multi-board Zonal Architecture environment using AI-G, D3-G, and VCP-G development boards.
- Practiced GPIO, CAN, I2C, and SPI interfacing, and explored FreeRTOS-based peripheral control.
- Gained hands-on experience in AI model training, fine-tuning, conversion, quantization, and NPU inference pipelines.

Nextchip

- Set up an ARM-based Embedded Linux development environment using cross-compilation, NFS, and UART.
- Studied GNU toolchains, Makefiles, static and shared libraries, and low-level file I/O.
- Practiced Linux system programming, including process management, IPC, POSIX threads, and synchronization.

Google AI Professional Certificate | Google — Aug 2026

- Completed 7 courses covering prompt engineering, AI-assisted data analysis, research, and application development.
- Built 20+ AI-assisted artifacts and a custom AI solution, applying prompt optimization and AI output evaluation.

PROJECTS

PROJECT 1 | Educational Firmware

- Developed a bank-aware internal Flash driver using STM32Cube HAL and a 24-sector memory map, supporting sector-based erase, aligned halfword writes, and memory-mapped reads for non-volatile data storage.
- Implemented buffered USB CDC communication through callbacks and a 512-byte circular receive buffer, decoupling host packet reception from application-level processing while tracking buffer saturation.
- Designed a unified serial communication interface for USB CDC and DMA-based UART channels, enabling application code to handle multiple communication paths through shared read, write, and availability APIs.

PROJECTS CONTINUED

PROJECT 2 | PC Monitor

- Built a Python telemetry collector using psutil and PySerial to sample CPU and RAM utilization and calculate network throughput, enabling periodic delivery of PC performance data to an STM32 board.
- Implemented an interrupt-driven UART receiver with framed packets and FreeRTOS task signaling to parse CPU, RAM, and network metrics, moving data processing out of the interrupt callback and enabling event-driven system updates.
- Designed a fixed-size circular history buffer and dynamically scaled LCD charts to retain recent measurements without increasing memory usage, enabling live visualization of CPU, RAM, and network trends on the embedded display.

PROJECT 3 | Smart Highway ITS

- Implemented a stateful vehicle-counting module using ByteTrack IDs and bounded per-camera track history to count each vehicle once and maintain lane- and class-specific traffic statistics.
- Fused object-detection and segmentation outputs by validating each vehicle's ground-contact point against the drivable-area mask, excluding vehicles outside the target roadway from the count.
- Analyzed tracking failures affecting small and distant vehicles, identified unstable track IDs as a source of duplicate counts, and proposed track confirmation, lost-track recovery, and distance-aware matching.

WORK EXPERIENCE

Hubmeca | Technical Support and Operations Specialist

Sep 2022 - Apr 2026

- Managed enterprise client data uploads and coordinated responses to system integration issues.
- Standardized workflows for 30-100 daily requests to prevent missed or duplicate processing.
- Coordinated 150-200 monthly data uploads and proactively addressed operational risks.

SKILLS

Programming: C, C++, Python

Embedded and system: FreeRTOS, UART, USB CDC, NPU post-processing, ByteTrack, image segmentation, OpenGL ES, V4L2

Platforms and tools: STM32CubeIDE, STM32CubeMX, STM32 HAL, STM32F4/F7, Embedded Linux, Nextchip APACHE6 SoM, Git, GitHub, Ubuntu, WSL2, Oracle VM VirtualBox, MobaXterm